Operating Systems - CS(D)

Quiz No. 7

Consider a demand-paged virtual memory system with a **24-bit** addressing scheme, **8KB page size,** and **8 MB of main memory.** Assume the access time for main memory is **100 ns** and the page fault service time is **10ms.** Assume the page table is stored in main memory.

1. If no TLB is used, how long does it take for a userprogram to load a byte from main memory into a CPU register? Assume the desired page is already in main memory.
   [3 marks]

2. Suppose we add a TLB to the system to speed up logical-to-physical address translation. With a TLB **access time of 10 ns** and a **TLB hit rate of 0.9**, what is the effective memory access time? Assume we only do a main-memory page table lookup after a TLB miss, and that the desired pages are always in main memory.
   [4 Marks]

3. With a TLB, what is the time required to load the page into memory **if there is a page fault.**
   [3 Marks]